

AHMED AYMAN ELSAWY

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer who takes AI systems from raw data to deployed product. Built a multi-agent RAG decision-support platform with a 96.6% accuracy risk classifier and a FastAPI triage service that auto-resolves ~85% of routine cases at zero LLM cost, an A+ AutoML graduation platform for non-technical users, and a bilingual mental health app powered by a Falcon-7B LoRA fine-tune. Strong across data engineering, model evaluation, agent orchestration (LangGraph), and Dockerized deployment. Bachelor of Artificial Intelligence (2026).

EDUCATION

Delta University for Science and Technology Oct 2022 - Jun 2026
Bachelor of Artificial Intelligence
Graduation Project Grade: A+
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Data Structures and Algorithms

EXPERIENCE

Generative and Agentic AI Trainee - Digital Egypt Pioneers Initiative (DEPI) Nov 2025 - Jun 2026

- Completed intensive training covering Generative AI, RAG systems, and multi-agent architectures using Transformers and prompt engineering.
- Implemented LoRA fine-tuning pipelines and tool-calling AI agents with memory systems and autonomous workflows.
- Built and deployed end-to-end AI solutions involving retrieval-augmented generation and agentic decision loops.

PROJECTS

SmartML - Intelligent AutoML Platform for Small Datasets | [GitHub](#) Graduation Project (A+)
Python, FastAPI, React, Docker, PyTorch, Scikit-learn, XGBoost, LightGBM, PostgreSQL, LangGraph, OpenAI API, Redis

- Built an end-to-end ML platform that guides users from dataset analysis to model deployment through AI-assisted recommendations and automated workflows.
- Implemented data quality optimization, synthetic data generation, feature engineering, and leakage-safe AutoML pipelines to improve model reliability.
- Delivered a full-stack pipeline with an interactive UI that lets non-technical users upload data, configure models, and export deployment-ready artifacts.

PHARAOH-Guard - AI Crowd-Risk and Heritage Protection System | [GitHub](#) DEPI Graduation Project
Python, Scikit-learn, LangChain, LangGraph, FastAPI, FAISS, Groq LLM, SQLite

- Built an end-to-end decision-support system for crowd safety across 8 Egyptian heritage sites: risk prediction, operational guidance, and automated dispatch with audit logging.
- Trained a 4-class risk classifier on a 55K-record engineered dataset (39 leakage-controlled features), achieving 96.6% accuracy and 0.92 macro-F1 with 98% recall on the rare Critical class.
- Designed a RAG and multi-agent architecture (LangGraph supervisor team) over a 5,852-document FAISS archive, with a deterministic safety core: the LLM decides when to act, never what the actions are.
- Shipped a FastAPI service with staged triage that auto-closes ~85% of routine cases at zero LLM cost, plus a SQLite audit trail and a real-time operations dashboard.

Intelligent University Timetable Generator | [GitHub](#) 2026
Python, Genetic Algorithms, Google OR-Tools (CP-SAT), Streamlit, Pandas, OpenPyXL

- Designed a hybrid scheduling system (Greedy heuristic + CP-SAT via Google OR-Tools + Genetic Algorithm) that auto-generates conflict-free university timetables.
- Schedules 180+ student groups and 40+ instructors across 6 days in under 5 minutes, a task that takes human schedulers 1-3 days.
- Built an interactive Streamlit dashboard with dynamic table views and export controls.

Mental Health Chatbot | [GitHub](#) 2025
Python, Flask, NLP, Hugging Face Transformers, LoRA

- Built a bilingual (Arabic/English) full-stack web app for university student mental health check-ins.
- Fine-tuned Falcon-7B with LoRA adapters to generate personalized responses from a 13-question student assessment.
- Implemented end-to-end data security: Fernet encryption for personal fields and PBKDF2-SHA256 password hashing with unique salts.

Python, TensorFlow, Keras, EfficientNetB6, OpenCV, tf.data, Matplotlib

- Developed an end-to-end binary CNN classifier detecting malignant melanoma from 10,000+ Kaggle dermoscopy images (180x180 RGB, 80/20 train/validation split), trained with Adam (lr=0.001) and binary cross-entropy loss.
- Applied transfer learning with a frozen ImageNet-pretrained EfficientNetB6 backbone feeding a custom dense head (512, 256, and 128 units with 0.5/0.3 dropout) and a sigmoid output for probability-scored predictions.
- Engineered an optimized tf.data input pipeline (automated Kaggle API ingestion, caching, shuffling, AUTOTUNE prefetching) with on-the-fly augmentation and EarlyStopping (patience=3, best-weights restore) to prevent overfitting.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Class-Imbalance Handling

Deep Learning and CV: TensorFlow, PyTorch, Keras, CNNs, Transfer Learning, OpenCV

GenAI and LLMs: RAG Pipelines, Vector Databases (FAISS), Embeddings, LoRA Fine-tuning, Prompt Engineering, Agentic AI

Frameworks and APIs: LangChain, LangGraph, Hugging Face Transformers, OpenAI API

Backend and Deployment: FastAPI, Flask, REST APIs, Docker, Model Deployment, MLOps

Data and Databases: NumPy, Pandas, Data Preprocessing, PostgreSQL, MySQL, MongoDB

Tools: Git, GitHub, Jupyter Notebook, Google Colab, Streamlit

CERTIFICATIONS

- [Introduction to Artificial Intelligence \(AI\)](#) - Coursera
- [Mathematics for Machine Learning: Linear Algebra](#) - Coursera
- [Building LLM Applications with Prompt Engineering](#) - NVIDIA (2024)